Ranking and price valuation: two-code comparison

Ptixiaki versus the attached nineframes-eiv-worker archive. Only tree-based price estimates and their input data are compared.

Conclusion: same two-stage XGBoost design, but materially different targets, data, geography and features. Similar euro estimates are not assured.

What is similar

• Both train a first tree, form residual = (price - prediction) / price × 100, average nearby residuals as GeoPremium, then train a second tree with GeoPremium.

• Both use histogram XGBoost with depth 8, min-child-weight 30, 500 trees, learning rate 0.05 and 0.8 row/feature sampling; both export ONNX and ultimately report price per m².

Differences that change the estimate

Topic	Ptixiaki	Archive ranking
Price target	Fits asking EUR/m² directly.	Adjusts scraped asking by area premium; sell training also includes eligible verified sale prices. Fits log1p(EUR/m²), then uses expm1 at prediction.
Verified sales	No government-sale union in Worker tree training.	Pools eligible verified sales with listings at equal nominal row weight, despite many missing attributes. Sell models only.
Feature contract	Engineered fields before fitting; residence Phase2 checks exact 30-feature order and finite values.	Trains on raw SQL fields, but Go predicts with engineered fields. Missing manifest names become NaN; count checks do not detect this mismatch.
GeoPremium	Up to 50 phase-one residual neighbors within 300 m; new property gets a fresh mean.	Up to 50 within 160 m; for a new property, copies one nearby saved premium or uses zero. Verified-sale rows get zero.
Training pool	Residence query needs no valuations row; land/commercial do. Requires positive property asking price.	All scraped segments need a valuations row. Sell listings also need valuations.asking_price > €1,000; the target still uses properties.asking_price.
Other inputs	Residence includes Elevator; admin boundary level 6; fixed 2026 age logic.	Residence SQL omits Elevator; picks an admin boundary from levels 6-9; Go age logic uses current year.

Both do an 80/20 diagnostic split and refit the final tree on all training rows. Ptixiaki records validation R² and fixes the model seed; the archive reports median percentage errors and does not set a model random_state explicitly.

Verified sales: the important caution

The archive appends government sale rows with UNION ALL. It uses verified_sale_price / main_area_sqm as the target and does not apply the scraped-listing premium correction to these rows. Only eligible sale rows enter; the separate held-out government cohort is excluded.

Most property details are missing

For residences, the archive explicitly inserts NULL for facade, renovation, heating, view, bathrooms, floor, bedrooms, parking, pool, garden and other fields. Missing construction year becomes 0. Most raw numeric/text-array fields become NaN during Python conversion; missing categorical values become code -1. XGBoost can fit these rows, but it cannot learn the effects of features the sale records never contain.

Methodological concern: putting these sparse verified-sale rows and detailed listings into one tree at equal row weight, without source-aware handling of missingness, source weighting or a demonstrated matched-property backtest, is questionable. It may help calibrate prices by area/size, but it can also teach source-specific missing-value patterns. The code does not prove a net benefit.

How the archive uses each verified sale

• Its known area, subtype, size and sale price can influence the fitted trees. The sale is not reduced in weight because its other fields are NULL.

• The IsVerifiedSale flag is dropped as a tree feature. Nevertheless, the many NULL fields and construction-year 0 can act as an indirect source marker.

• Its synthetic gov_ UID is absent from the property_geo_data neighbor search, so its training GeoPremium is zero and it does not contribute as a spatial neighbor there.

What the valuations row does

For scraped listings, the archive uses INNER JOIN valuations v ON p.uid = v.property_uid. A missing row excludes the listing. On sale listings, v.asking_price > €1,000 is an additional filter. No other valuations column is fed to the tree: the price target comes from properties.asking_price. Verified-sale UNION rows bypass this join.

Interpretation

The differences can be large, but the code cannot establish a fixed percentage or which model is consistently higher. The archive ZIP has no trained price artifacts for a matched-property comparison. No project source, trained model or database record was changed.

Source trace. Ptixiaki: Worker/Models/TreeCode/Tree.py (181-195, 671-714); Worker/training/GeoPremium.go (92-105); Worker/testing/GetGeoPremium.go (21-83); Worker/database/QueryBuilder.go (37-61, 131-170); ml_shared/explain_artifacts.py (25-56). Archive: internal/ranking/training/train.py (120-283, 318-412); internal/ranking/rules.go (128-155, 194-289, 293-409); configs.go (20-63); features.go (42-134); geopremium.go (43-84); encoder.go (26-88).